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AI Course

Capstone Project Action Plan

For students (instructor's review required)

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Course	**AI Course**
Team Name	SIC-UTT Medical AI Innovators
Team Leader / Members	**Le Dinh Duy** (Team Leader) Tran Van B, Nguyen Thi C, Pham Van D, Hoang Thi E
Project Title	**Multimodal Medical AI Assistant: Real-time Clinical Triage and Automated Lab Report Diagnostic System**

Goal

- Develop an enterprise-grade Multimodal Medical AI Chatbot designed to provide real-time clinical triage, specialty routing, and automated laboratory report interpretation.
- Integrate multi-input modalities: Vietnamese Speech-to-Text (Faster-Whisper), Complete Blood Count (CBC) document OCR with biochemical sanity range verification, and fine-tuned PhoBERT Medical Named Entity Recognition (NER).
- Build a Hybrid Late Fusion diagnostic framework ($P_{\text{final}} = \alpha P_{\text{tabular}} + (1-\alpha)P_{\text{text}}$), a sub-0.5s Red Flag emergency triage engine, and an active entropy-driven clarification dialog loop to resolve diagnostic ambiguity.

Abstract

Hospital overcrowding, primary care bottlenecks, and delayed recognition of critical medical emergencies remain major challenges in contemporary healthcare. This project develops the **Multimodal Medical AI Assistant**, an end-to-end clinical platform structured as an enterprise modular monorepo:

- **Multimodal Ingestion**: Accommodates conversational text, streaming speech input via Faster-Whisper, and optical character recognition (OCR) of complete blood count (CBC) lab test panels with reference range validation.
- **AI Core**: Fine-tunes `vinai/phobert-base-v2` for medical entity recognition (symptoms, diseases), trains a GPU-accelerated `XGBoost` classifier over tabular biochemical indices, and unites them via Late Fusion to attain high diagnostic precision ($F_1 > 90\%$).
- **Interactive Ecosystem**: High-concurrency FastAPI server with WebSocket streaming typewriter, Next.js 14 dynamic side dashboard, and Qdrant vector database indexing Vietnam Ministry of Health ICD-10 clinical guidelines.

Method

1. **Multimodal Preprocessing & Speech/OCR Pipeline**: Deploy Faster-Whisper for low-latency Vietnamese voice transcription; utilize Tesseract/EasyOCR combined with regex pattern matchers and domain lexicons to parse WBC, RBC, HGB, PLT, and reference ranges.
2. **Medical NER & BIO Tagging**: Fine-tune PhoBERT Token Classification over Vietnamese clinical text datasets to extract entities labeled with `B-SYMPTOM`, `I-SYMPTOM`, `B-DISEASE`, `I-DISEASE`.
3. **Hybrid Late Fusion & Clarification Engine**: Formulate ensemble inference: $P_{\text{final}} = 0.45P_{\text{tabular}} + 0.55P_{\text{text}}$. When maximum confidence $P_{\max} < 0.70$, the system generates targeted clarification questions based on entropy minimization across differential diagnoses.
4. **Medical RAG Knowledge Base**: Vectorize ICD-10 ontologies and treatment protocols using `BAAI/bge-m3` embeddings indexed in Qdrant, providing factually grounded, clinically safe guidance.

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Data

- **Clinical Conversation & Text Datasets**: Curated from `ViMedical_Disease` and `medical-vietnamese-qa`, encompassing over 15,000 annotated doctor-patient dialogues with BIO token tags for medical entities and symptom descriptions.
- **Complete Blood Count (CBC) Tabular Dataset**: Constructed benchmark synthetic dataset covering 12 core biochemical parameters (WBC, RBC, HGB, HCT, MCV, MCH, MCHC, PLT, LYM, NEU, GLU, CREAT) across 10 prevalent diagnostic classes (Dengue hemorrhagic fever, bacterial infection, anemia, diabetes, thrombocytopenia, etc.).
- **Medical Ontologies & Expert Lexicons**: Official Vietnamese ICD-10 clinical diagnosis guidelines, acute Red Flag criteria (chest pain, dyspnea, stroke indicators), and a suite of 50 verified end-to-end clinical acceptance scenarios.

Expected Outcome

- **Quantitative Technical Milestones**:
  - PhoBERT Medical NER achieving Macro $F_1 \geq 88.5\%$.
  - Hybrid Late Fusion diagnostic classifier reaching multi-class $F_1 \geq 90.2\%$.
  - Red Flag emergency detection triggering within ≤ 0.50 seconds with 100% critical recall.
  - Blood test OCR parameter extraction accuracy exceeding 93.0%.
- **Integrated Enterprise Deliverables**: Production-ready containerized microservices (Docker-compose), responsive Next.js 14 frontend, low-latency WebSocket streaming, and administrative audit portal.
- **Healthcare & Societal Benefits**: Enables preliminary pre-hospital risk triage for households, immediate life-saving emergency escalation (115 hotline dispatch), and substantial workload reduction for emergency departments.

Role by Member

Le Dinh Duy (Leader / AI Architect)	Overall monorepo architecture; PhoBERT Medical NER fine-tuning; Hybrid Late Fusion algorithm design; entropy-based clarification logic; FastAPI core orchestration.
Tran Van B (ML Engineer)	Clinical tabular CBC data engineering; XGBoost multi-class classifier optimization; reference range sanity validation logic; cross-validation benchmarking.
Nguyen Thi C (Vision & Speech Eng.)	Faster-Whisper audio transcription integration; lab report OCR extraction pipeline; regular expression normalization; preprocessing unit test coverage.
Pham Van D (RAG & Knowledge Eng.)	Qdrant vector database infrastructure; BAAI/bge-m3 dense semantic embeddings; ICD-10 medical ontology alignment; 50 clinical evaluation test cases.
Hoang Thi E (Fullstack & DevOps)	Next.js 14 web client development; dynamic medical side dashboard; WebSocket typewriter streaming; Docker-compose orchestration and automated CI/CD pipeline.

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Schedule Summary

Week	Work Package	Implementation Details and Key Milestones	Lead
Week 1	Project Inception & Architecture	Requirements engineering, WBS scheduling, Docker monorepo setup	All
Week 2	Data Ingestion & Lexicons	ViMedical corpus labeling, synthetic CBC dataset, ICD-10 database	B, C, D
Week 3	Core AI Model Training	PhoBERT NER fine-tuning, XGBoost clinical tabular model, NLP classifier	Duy, B
Week 4	Multimodal Ingestion & RAG	Faster-Whisper STT service, lab test OCR pipeline, Qdrant BGE-M3 RAG	C, D
Week 5	Hybrid Late Fusion Engine	Late fusion optimization, sub-0.5s Red Flag triage, clarification loop	Duy, D
Week 6	Backend API & Frontend UI	FastAPI WebSocket streaming, Next.js 14 medical dashboard	Duy, E
Week 7	System Integration & Testing	End-to-end execution of 50 clinical test cases, stress and latency tests	All
Week 8	Final Report & Delivery	Final report writing, slide presentation, artifact packaging and demo	All

Comment & Assessment

Instructor Review and Constructive Feedback:

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Instructor Signature:

Evaluation Date:.....